dotdraw

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```
This file is part of CasADi.

CasADi -- A symbolic framework for dynamic optimization.
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```


[1]:
```
import casadi as ca
from casadi.tools import *
```

An SX graph

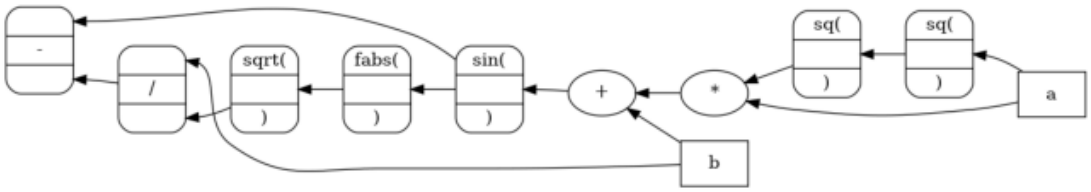
[2]:
```
a = ca.SX.sym("a")
b = ca.SX.sym("b")
```

[3]:
```
c = ca.sin(a**5 + b)
```

[4]:
```
c = c - b/ ca.sqrt(ca.fabs(c))
print(c)
```

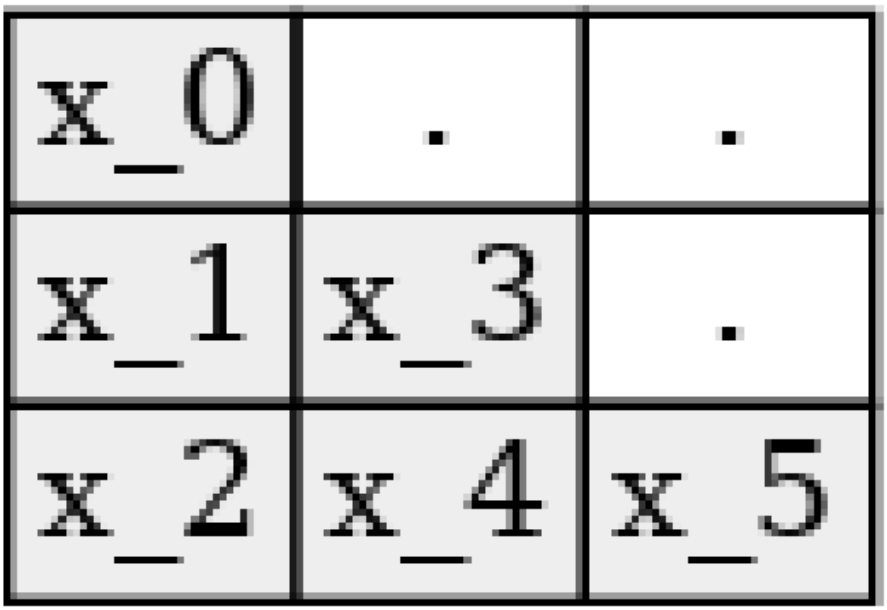
```
@1=sin(((a*sq(sq(a)))+b)), (@1-(b/sqrt(fabs(@1))))
```

[5]:
```
dotdraw(c)
```

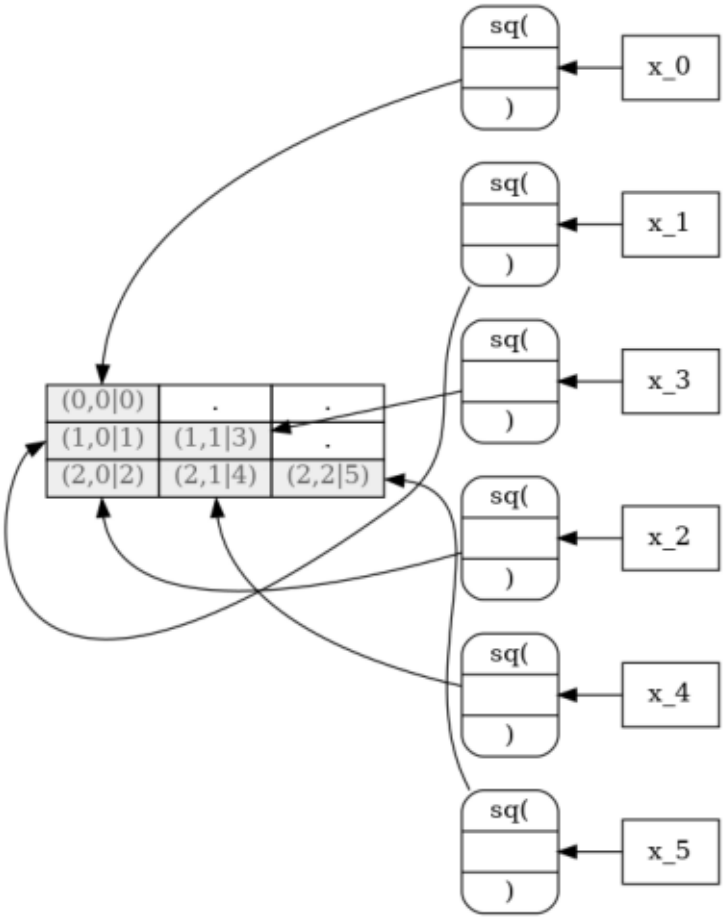


An SX

[6]:
```
dotdraw(ca.SX.sym("x",ca.Sparsity.lower(3)))
```



[7]:
```
dotdraw(ca.SX.sym("x",ca.Sparsity.lower(3))**2)
```



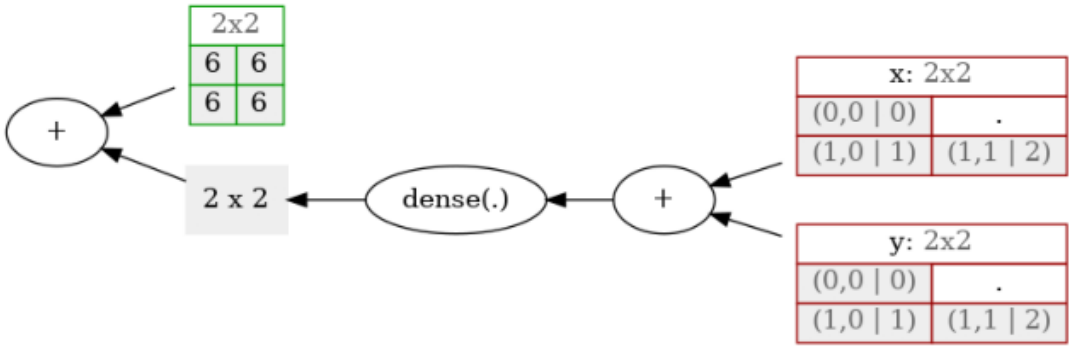
An MX graph

[8]:
```
x = ca.MX.sym("x",ca.Sparsity.lower(2))
y = ca.MX.sym("y",ca.Sparsity.lower(2))
```

[9]:
```
z = ca.MX.sym("z",4,2)
```

[10]:
```
zz = x+y+6
```

[11]:
```
dotdraw(zz)
```



[12]:
```
f = ca.Function("magic", [z,y],[z+x[0,0],x-y],{"allow_free":True})
```

[13]:
```
z,z2 = f(ca.vertcat(x,y),zz.T)
```

[14]:
```
z = z[:2,:] +x + ca.cos(x) - ca.sin(x) / ca.tan(z2)
```

[15]:
```
dotdraw(z)
```

